THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 3: **Pair of Linear Equations in Two Variables** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** infinitely many solutions	$\frac{1}{2}=\frac{2}{4}=\frac{3}{6}$ ⇒ coincident lines
2	**(b)** parallel	$\frac{2}{4}=\frac{3}{6}\neq\frac{7}{9}$
3	**(b)** $k\neq 6$	$\frac{k}{3}\neq\frac{2}{1}\Rightarrow k\neq 6$
4	**(a)** $(3,2)$	$2(3)+3(2)=12$ ✓
5	**(b)** Incorrect — the pair has infinitely many solutions and is consistent	Coincident lines ⇒ dependent and consistent
6	**(b)** $(3,2)$	Adding: $2x=6\Rightarrow x=3,\ y=2$
7	**(b)** Student B is correct	$\frac{3}{9}=\frac{-1}{-3}=\frac{3}{9}$ ⇒ coincident
8	**(c)** 3	$4+k=7\Rightarrow k=3$
9	**(b)** parallel to the y-axis	All points have $x=4$
10	**(a)** $12, 8$	$x+y=20,\ x-y=4\Rightarrow 12, 8$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$x=14-y\Rightarrow 14-2y=4\Rightarrow y=5,\ x=9$	1+1
12	$\frac{1}{3}=\frac{2}{k}\Rightarrow k=6$; then $\frac{c_1}{c_2}=\frac{-5}{15}=-\frac{1}{3}\neq\frac{1}{3}$ ✓. So $k=6$	1+1
13	$\frac{a_1}{a_2}=\frac{3/2}{9}=\frac{1}{6}$, $\frac{b_1}{b_2}=\frac{5/3}{-10}=-\frac{1}{6}$; unequal ⇒ unique solution ⇒ **consistent**	1+1
14	$l+w=36,\ l-w=4\Rightarrow l=20$ m, $w=16$ m	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Multiply 2nd by 2: $4x-4y=4$; add: $7x=14\Rightarrow x=2$; then $y=1$	1+1+1
16	$5x+7y=50,\ 7x+5y=46$. Adding: $x+y=8$; subtracting: $y-x=2$ ⇒ pencil ₹3, pen ₹5	1+1+1
17	$j+5=3(s+5),\ j-5=7(s-5)\Rightarrow j=3s+10,\ 3s+5=7s-35\Rightarrow s=10,\ j=40$ years	1+1+1
	OR: $3(x-1)=y,\ 4x=y+8\Rightarrow 4x=3x+5\Rightarrow x=5,\ y=12$. Fraction $\frac{5}{12}$	1+1+1
18	Lines meet at $(2,3)$; x-intercepts $(-1,0)$ and $(4,0)$. Vertices: $(2,3),(-1,0),(4,0)$	2+1
	OR: Points: $(0,2),(6,0)$ and $(0,-4),(6,0)$. Lines meet at $(6,0)\Rightarrow x=6,\ y=0$	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $x+10y=105,\ x+15y=155$ (ii) $5y=50\Rightarrow y=$ ₹10, $x=$ ₹5 (iii) $5+250=$ ₹255 (iv) False — the fixed charge ₹5 is payable	1+2+1+1
	OR: $x+20y=1000,\ x+26y=1180\Rightarrow 6y=180,\ y=$ ₹30, $x=$ ₹400. For 30 days: $400+900=$ ₹1300	2+2+1
20	Let number $=10x+y$: $x+y=9$; $9(10x+y)=2(10y+x)\Rightarrow 88x=11y\Rightarrow y=8x\Rightarrow x=1,\ y=8$. Number 18. Verify: $9\times 18=162=2\times 81$ ✓	2+2+1
	OR: (i) $x+y=25,\ 50x+100y=2000\Rightarrow y=15$ (₹100), $x=10$ (₹50) (ii) $x+y=30,\ x+2y=40\Rightarrow$ 20 notes of ₹50 and 10 of ₹100	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 3: **Pair of Linear Equations in Two Variables** | Class 10 Mathematics | Maximum Marks: 40

SECTION — A (Objective Type) — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** intersecting	$\frac{1}{3} \neq \frac{-2}{4}$
2	**(d)** infinitely many	$\frac{2}{4} = \frac{3}{6} = \frac{9}{18}$
3	**(a)** 10	$\frac{1}{5} = \frac{2}{k} \Rightarrow k = 10$; $\frac{3}{-7} \neq \frac{1}{5}$
4	**(a)** $(3, 1)$	Adding: $2x = 6$
5	**(b)** Incorrect — parallel lines give no solution, so the pair is inconsistent	Parallel lines ⇒ no common point
6	**(b)** $(2, 3)$	Common point $(2, 3)$
7	**(a)** Student A is correct	$3 + 4 = 7,\ 6 - 4 = 2$ ✓; for B: $8 - 3 = 5 \neq 2$
8	**(a)** $\frac{15}{4}$	$\frac{3}{2} = \frac{2k}{5} \Rightarrow k = \frac{15}{4}$
9	**(d)** no solution	Two distinct lines parallel to the x-axis
10	**(a)** $24, 12$	$x = 2y,\ 3y = 36$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$3x + 3y = 15$; adding with 2nd: $5x = 19 \Rightarrow x = \frac{19}{5},\ y = \frac{6}{5}$	1+1
12	$\frac{5}{7} \neq \frac{-4}{6}$ ⇒ unique solution ⇒ **consistent**	1+1
13	$\frac{2}{a-b} = \frac{3}{a+b} \Rightarrow a = 5b$; $\frac{3}{a+b} = \frac{7}{3a+b-2} \Rightarrow 2a - 4b = 6 \Rightarrow b = 1,\ a = 5$	1+1
14	$x - y = 26,\ x = 3y \Rightarrow 2y = 26 \Rightarrow y = 13,\ x = 39$	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$2x + 3y = 13,\ 4x + 5y = 23$. $x = \frac{13-3y}{2} \Rightarrow 26 - 6y + 5y = 23 \Rightarrow y = 3,\ x = 2$	1+1+1
16	Lines meet at $(1, 4)$; x-intercepts $(3, 0)$ and $(-1, 0)$. Vertices $(1, 4), (3, 0), (-1, 0)$ (area $= \frac{1}{2} \times 4 \times 4 = 8$ sq units)	2+1
17	$x + 4y = 27,\ x + 2y = 21 \Rightarrow y =$ ₹3, $x =$ ₹15	1+1+1
	OR: $f - 2 = 5(s - 2),\ f + 2 = 3(s + 2) + 8 \Rightarrow 5s - 8 = 3s + 12 \Rightarrow s = 10,\ f = 42$ years	1+1+1
18	$5x - 5y = 100 \Rightarrow x - y = 20$; $x + y = 100 \Rightarrow$ 60 km/h and 40 km/h	1+1+1
	OR: $x + y = 180,\ x - y = 18 \Rightarrow 99^\circ$ and 81°	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $3x - y = 40,\ 4x - 2y = 50$ (ii) $2x - y = 25 \Rightarrow x = 15,\ y = 5$ (iii) $15 + 5 = 20$ questions	2+2+1
	OR: $1.08x + 0.9y = 1008,\ 1.1x + 0.92y = 1028 \Rightarrow x =$ ₹600, $y =$ ₹400. Check: $648 + 360 = 1008,\ 660 + 368 = 1028$ ✓	2+2+1
20	$(x - 5)(y + 3) = xy - 9 \Rightarrow 3x - 5y = 6$; $(x + 3)(y + 2) = xy + 67 \Rightarrow 2x + 3y = 61 \Rightarrow x = 17,\ y = 9$. Verify: $12 \times 12 = 144 = 153 - 9,\ 20 \times 11 = 220 = 153 + 67$ ✓	2+2+1
	OR: $(v + 10)(t - 2) = vt \Rightarrow -v + 5t = 10$; $(v - 10)(t + 3) = vt \Rightarrow 3v - 10t = 30 \Rightarrow t = 12$ h, $v = 50$ km/h; distance $= 600$ km	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.